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AI-DRIVEN CYBER SECURITY

AN AI AGENT CAN DO MUCH MORE THAN ANSWER QUESTIONS. IT CAN EXECUTE COMMANDS, ACCESS FILES, CALL APIS, AND MODIFY SYSTEMS. BUT WHAT HAPPENS WHEN IT USES THESE POWERS IN THE WRONG WAY?

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WHAT IS TOOL & PERMISSION ABUSE?

- Tool abuse occurs when an AI agent misuses its available tools.
- Permission abuse occurs when an agent has or uses more access than necessary.
- Example: A security agent deletes logs instead of simply analyzing them.

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EXCESSIVE PERMISSIONS

- Giving an AI agent unnecessary access increases security risks.
- Follow the Principle of Least Privilege.
- Give agents only the permissions required for their specific tasks.

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DIRECT EXECUTION RISKS

- Agents with shell, file-system, or API access can cause real damage.
- Unvalidated commands may delete files, expose secrets, or modify systems.
- Never execute AI-generated commands blindly.

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PROMPT INJECTION AND TOOL ABUSE

- Malicious instructions in user input or external documents may manipulate an agent.
- A manipulated agent might misuse tools or attempt unauthorized actions.
- Treat external content as untrusted and enforce security controls outside the AI model.

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SECURITY GUARDRAILS

- Allowlist approved tools and restrict their capabilities.
- Validate tool arguments before execution.
- Block dangerous commands and unauthorized file or network access.
- Require human approval for sensitive actions.

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MONITORING AND HUMAN OVERSIGHT

- Log tool calls, arguments, results, and denied actions.
- Set execution limits and timeouts.
- Use a sandbox for risky operations.
- Regularly test agents against malicious inputs.